

Foundation (Jalapeno)

Q1. Simplify $3 \cdot \sqrt{2} + 2 \cdot \sqrt{2}$

- (A) $5 \cdot \sqrt{2}$
- (B) $5 \cdot \sqrt{3}$
- (C) $\sqrt{2}$
- (D) $\sqrt{3}$
- (E) $2 \cdot \sqrt{3}$

Q2. Simplify $\sqrt{16} - \sqrt{9}$

- (A) 3
- (B) 4
- (C) 1
- (D) -1
- (E) -3

Q3. Simplify $\sqrt{8} \cdot \sqrt{2}$

- (A) 4
- (B) 8
- (C) 2
- (D) $\sqrt{16}$
- (E) $\sqrt{4}$

Q4. Simplify $\frac{\sqrt{16}}{\sqrt{4}}$

- (A) 2
- (B) 4
- (C) 1
- (D) $\sqrt{4}$
- (E) $\sqrt{16}$

Q5. Simplify $\sqrt{20}$

- (A) $2\sqrt{5}$
- (B) $\sqrt{4} \cdot \sqrt{5}$
- (C) $4\sqrt{5}$
- (D) $\sqrt{2} \cdot \sqrt{10}$
- (E) $2 \cdot \sqrt{10}$

Q6. Simplify $\sqrt{32}$

- (A) $4\sqrt{2}$
- (B) $2\sqrt{8}$
- (C) 8
- (D) $\sqrt{16} \cdot \sqrt{2}$
- (E) $8\sqrt{2}$

Q7. Simplify $\frac{\sqrt{48}}{\sqrt{3}}$

- (A) 4
- (B) $4\sqrt{3}$
- (C) $\sqrt{16}$
- (D) $\sqrt{16} \cdot \sqrt{3}$
- (E) $4\sqrt{4}$

Q8. Simplify $\sqrt{8} \cdot \sqrt{50}$

- (A) 20
- (B) 40
- (C) $\sqrt{400}$
- (D) $20\sqrt{2}$
- (E) $10\sqrt{8}$

Open Ended Questions (FRQ)

Q9. Simplify the expression: $\frac{\sqrt{96}}{\sqrt{6}}$

Q10. Simplify the expression: $2\sqrt{18} + 5\sqrt{8}$

Chef's Tips

- Make sure to check the student's simplification process.
- Students should be able to provide the correct simplified form of the given expressions.
- Remind students to apply the rules of radicals to their work.